

RNA Tumour Viruses

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INTRODUCTION TO RNA TUMOUR VIRUSES

Worldwide, it is estimated that 15% of cancer cases are aetiologically linked to viral infection, accounting for nearly 1.5 million new cases and 900 000 deaths annually (Butel, 2000). For most human cancers, the precise cause of tumorigenesis is unknown, although epidemiologic studies have linked infection by specific DNA- and RNA-containing viruses to the development of particular cancers. In viral-associated human cancers, genomic instability and subsequent multi-step tumorigenesis is associated with the expression of viral oncoproteins. Thus, oncogenic viruses provide an opportunity to directly investigate molecular and cellular mechanisms that lead to cancer. This chapter will focus predominantly on RNA viruses that cause cancer in humans.

Human RNA Tumour Viruses

Of the human RNA viruses, human T-cell leukaemia virus type-I (HTLV-I) and hepatitis type-C virus (HCV) of the *retroviridae* and *flaviviridae* families, respectively, have been established as transforming viruses. HTLV-I is the aetiological agent of adult T-cell leukaemia (ATL), a disease characterized by malignant proliferation of CD4⁺ T lymphocytes that develops in a small percentage of HTLV-I-infected individuals (Kondo *et al.*, 1987). Although acute HCV infection often causes only mild illness, it has become increasingly evident that HCV is responsible for substantial morbidity and mortality, particularly from chronic liver disease and hepatocellular carcinoma (Liang *et al.*, 2004) (see **Liver, Gallbladder, and Extrahepatic Bile Ducts**). Here, we will discuss retroviruses as the prototypic representatives of the RNA tumour viruses. Animal retroviruses induce cellular transformation via several distinct mechanisms: transduction of proto-oncogenes, proviral integration adjacent to proto-oncogenes, trans-activation or repression of cellular gene expression, and disruption of cell cycle control and DNA

repair, each of which promote viral efforts to drive replication of the proviral genome.

Overview of Retroviral Replication

Retroviruses bind to specific cellular receptors and following internalization and uncoating, the viral-encoded reverse transcriptase transcribes the viral genomic RNA into single-stranded complementary deoxyribonucleic acid (cDNA) and subsequently into double-stranded deoxyribonucleic acid (dsDNA) (Figure 1). After completing dsDNA synthesis, the viral DNA translocates to the nucleus, and integrates, usually randomly, into the host chromosomal DNA. The integrated viral genome, also known as the *provirus*, then replicates during cellular division. The retroviral genome encodes the structural and enzymatic genes *gag*, *pro*, *pol*, and *env* and is flanked on the 5' and 3' ends by non-coding long terminal repeats (LTRs) composed of U3, R, and U5 regions. The LTRs contain promoter elements and RNA processing signals that are necessary for viral gene expression. Cellular machinery is required for viral genes to be transcribed, spliced, and exported to the cytoplasm where they are translated into precursor proteins. Assembly of progeny virions from these precursor proteins occurs coincident with packaging of two copies of the single-stranded viral RNA genome. Virion budding and release from the cell is followed by the proteolytic cleavage of precursor proteins to form mature virions (reviewed in Goff, 2001).

Role of Retroviral Replication in Tumourigenesis

Retroviruses have been identified in almost all animal species and many have been shown to transform cells in culture and/or cause tumours in animals. Most transforming retroviruses mediate this effect through the action of cellular oncogenes. Transforming retroviruses fall into two groups: transducing retroviruses (e.g., Rous sarcoma virus), which carry a proto-oncogene within their genomes, and *cis*-activating retroviruses (e.g., avian leukosis virus and mouse mammary tumour virus), which lack a proto-oncogene, but

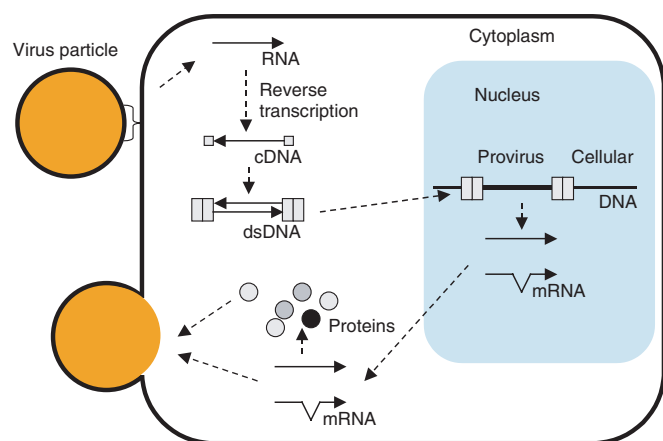


Figure 1 Retrovirus life cycle.

stimulate transformation by integrating in the vicinity of a cellular oncogene and activating the expression of nearby genes. A proto-oncogene is a normal cellular gene that, when altered by mutation or aberrant regulation, becomes an active oncogene. More than 30 oncogenes encoded by transducing retroviruses have been identified and more than 70 proto-oncogenes have been shown to be regulated by insertion of *cis*-activating retroviral proviruses (Rosenberg and Jolicoeur, 1997). In addition to expression or activation of oncogenes, viral regulatory proteins that function primarily to enhance transcription from the viral LTR, such as HTLV-I Tax, have been shown to dysregulate several cellular processes and lead to tumour formation (Yoshida, 2005).

HUMAN TUMOURIGENIC RETROVIRUSES: HTLV-I

HTLV-I, the first discovered pathogenic human retrovirus, is now recognized as the causative agent of ATL and tropical spastic paraparesis/human T-cell leukaemia virus type-I-associated myelopathy (TSP/HAM), a progressive neurological inflammatory syndrome (Gessain *et al.*, 1985). HTLV-I has also been associated with other clinical disorders including HTLV-I-associated arthropathy, HTLV-I-associated uveitis, infective dermatitis, and polymyositis (Buggage, 2003; Nicot, 2005; Watanabe, 1997).

Transmission of HTLV-I

The nature of HTLV-I transmission is an important factor in the development of ATL. The most efficient transmission of HTLV-I occurs from infected to uninfected cells via a virologic synapse (Igakura *et al.*, 2003). Cell contact induces cytoskeletal polarization of the infected cell at the cell–cell junction. Viral proteins and genomic RNA accumulate at the junction and are transferred to the target cell. The virus integrates randomly into host chromosomes and the HTLV-I provirus, in both asymptomatic viral carriers and ATL patients, is detected almost exclusively in CD4⁺ T

lymphocytes. However, other haematopoietic cells, including non-CD4⁺ T lymphocytes, B lymphocytes, monocytes, macrophages, and dendritic cells, also help maintain HTLV-I infection during the lifespan of an individual (reviewed in Manel *et al.*, 2005). Over the course of infection, expansion of a subset of infected CD4⁺ T lymphocytes containing one or a few clonally integrated HTLV-I proviruses adds to the population of oligoclonally infected cells (Eiraku *et al.*, 1998). Individual behaviours and exposure routes have been associated with transmission of HTLV-I: parenteral transmission by blood transfusion, mother-to-infant transmission through breast-feeding, and sexual intercourse. Each mode of transmission is distinct and the disease outcome appears to vary, depending on the route of transmission.

Transmission by Blood Transfusion

Intravenous routes, such as blood transfusion, appear to be the most efficient modes of HTLV-I transmission. Parenteral transmission of the virus from donor to recipient requires live infected lymphocytes (Kannagi *et al.*, 2004). In retrospective studies of transfusion patients who received HTLV-I-contaminated blood products, 40–60% were seropositive by 2 months post-transfusion (Manns *et al.*, 1992). There is also a high HTLV-I seroprevalence among intravenous drug users who share contaminated needles and syringes. Although blood-mediated transmission of HTLV-I-infected lymphocytes is an important risk factor for TSP/HAM, patients developing ATL following blood transfusion is rare (Gout *et al.*, 1990).

Mother-to-child Transmission

Mother-to-child transmission of HTLV-I occurs when an infant ingests infected milk-borne lymphocytes during prolonged breast-feeding. This route of transmission is associated with the development of ATL (Kannagi *et al.*, 2004). Twenty percent of children born of HTLV-I-infected women are seropositive. The risk of vertical transmission correlates with maternal proviral load and antibody titer (Ureta-Vidal *et al.*, 1999). After HTLV-I antibody screening was introduced as part of standard prenatal care in Japan, seropositive mothers were counselled to discontinue breast-feeding and the rate of infection in infants fed on artificial milk was significantly reduced (Hino *et al.*, 1997). The risk of infection by other routes of mother-to-child transmission, such as intrauterine or peri-partum, is low, possibly due to HTLV-I-induced apoptosis of placenta cells (Fujino *et al.*, 2000).

Sexual Transmission

Behaviours associated with sexual transmission of the virus parallel those of other sexually transmitted diseases: intercourse without condom use and having multiple or casual partners. The efficiency of sexual transmission is due to the presence of HTLV-I-infected lymphocytes in the semen of seropositive males. Female sexual partners of infected males often become HTLV-I sero-positive and the cross-sectional

studies indicate that in most endemic areas, male-to-female transmission occurs at a higher frequency than in the reverse (Roucoux *et al.*, 2005). However, female-to-male transmission may occur by transmission of infected cells through injured mucous membranes. As with viral transmission by blood transfusion, individuals infected with HTLV-I by sexual transmission have a lower risk of developing ATL than those infected by mother-to-child transmission.

Global Epidemiology of HTLV-I

Epidemiological aspects of HTLV-I have been thoroughly and continuously studied since the virus was first identified in 1980. Initial screens for virus infection typically utilize serological assays such as enzyme-linked immunosorbent assays (ELISA), western blotting, and indirect immunofluorescence to detect antibodies to viral proteins. In addition to testing for sero-conversion, polymerase chain reaction (PCR) assays are usually used as confirmatory assays to identify HTLV-I proviral DNA.

Geographic Prevalence

It is currently estimated that 10 to 20 million people worldwide are infected with HTLV-I (Edlich *et al.*, 2003). The virus is endemic in regions of southern Japan, Africa, Central and South America, the Caribbean, Eastern Europe, and the southeastern United States (Bangham, 2000). The sero-prevalence rate differs by geographic area, socio-demographic composition, and individual risk behaviours. For example, up to 10% of the general population in southwestern Japan is seropositive for HTLV-I, yet the sero-prevalence rate is as high as 37% in small, selected populations from this region (Yamaguchi, 1994; Mueller *et al.*, 1996). In addition, indicators of lower socio-economic status, such as education and income, are associated with HTLV-I infection. In non-endemic regions, such as North America and Europe, the virus is detected primarily in immigrants from endemic areas, intravenous drug users, and sex workers. Overall, the sero-prevalence of HTLV-I in non-endemic regions is low (e.g., <0.006% in Norway, France, and Greece and <0.03% in the United States and Canada) (Proietti *et al.*, 2005).

Individual Prevalence

HTLV-I sero-prevalence rates correlate strongly with the age and sex of an individual, with the greatest number of virus carriers being females, 40 to 50 years of age. The age-dependency is attributed to reduced chances of viral exposure in younger individuals. Reduced HTLV-I sero-prevalence in certain regions of the world has been attributed to modified sexual behaviour (increased condom use) and breast-feeding practices (increased popularity and feasibility of artificial milk). The 1.6-fold greater sero-prevalence rate in females can be attributed to the greater efficiency of male-to-female

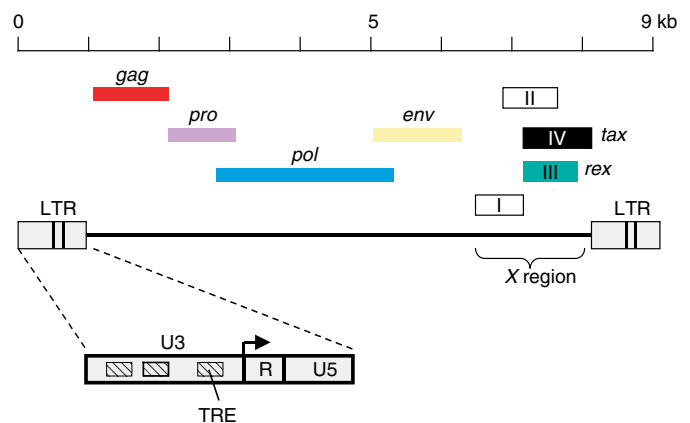


Figure 2 Genome structure of HTLV-I.

sexual transmission. Although there is an imbalance of sero-prevalence between males and females, the incidence of ATL is comparable in both sexes.

Viral Gene Expression

HTLV-I Genome

HTLV-I is an enveloped retrovirus with a diploid, positive-sense RNA genome that is 9032 nucleotides in length (Figure 2) (Seiki *et al.*, 1984). Like all known replication-competent retroviruses, the HTLV-I genome contains the gene sequence LTR-*gag-pol-env*-LTR. The presence of additional regulatory genes within a *pX* region located 3' of the *env* gene, characterizes HTLV-I as a complex retrovirus, distinct from other non-human retroviruses. The *X* region contains four open reading frames (ORFs), two of which [*rex* (ORF III) and *tax* (ORF IV)] encode regulatory proteins that are critical for HTLV-I replication and viral transforming ability. Rex is a 27-kDa protein that modulates the transport of unspliced and single-spliced viral RNA from the nucleus to the cytoplasm. By physically and temporally controlling the amount of viral RNA that is exported to the cytoplasm for translation, Rex can regulate the switch between latent and productive phases of the viral life cycle. The 40-kDa Tax protein is a potent trans-activator of viral gene expression. In addition, Tax activates the transcription of numerous cellular genes and is the major transforming protein of HTLV-I. Microarray analyses revealed over 300 genes that are upregulated in Tax-expressing cells (Pise-Masison *et al.*, 2002).

Transcriptional Regulation

HTLV-I contains a single viral promoter located within the U3 region of the LTR. The promoter contains three imperfectly conserved 21 base pair repeats called *Tax responsive elements (TREs)*. The TREs contain core sequences with homology to consensus cellular cyclic adenosine monophosphate-responsive elements (CREs) and are flanked by G/C rich sequences (Connor *et al.*, 2000; Cox *et al.*,

1995; Jeang *et al.*, 1988; Paca-uccaralertkun *et al.*, 1994). Importantly, Tax does not directly bind to DNA. Instead, Tax interacts with cellular transcription factors and coactivators to alter their DNA binding affinity and specificity. For example, Tax enhances the dimerization of cyclic adenosine monophosphate-response element binding protein (CREB) and increases its affinity for the imperfect viral CRE, but not the consensus cellular CRE (Adya *et al.*, 1994; Cohen-Fix *et al.*, 1996; Kwok *et al.*, 1996). The Tax-CREB-CRE complex creates a high affinity target site for recruitment of the cellular coactivators CREB binding protein (CBP) and its homologue, p300. The ability of Tax to directly interact with CREB and CBP/p300 allows this complex to bypass the normal cAMP signalling pathway and activate transcription from the viral LTR in the absence of CREB phosphorylation (Van Orden *et al.*, 2000).

RNA Processing

Synthesis of viral mRNA from an integrated proviral DNA results in a single primary transcript that can be directed to multiple pathways (Figure 3). A portion of these 8.5-kb transcripts is exported to the cytoplasm, where they are either packaged as genomes into progeny virions or are translated to Gag-Pro and Gag-Pro-Pol polyproteins. Alternatively, the transcript may be spliced in a Rex-dependent manner to form subgenomic mRNAs that encode the overlapping genes *env*, *rex*, and *tax*. *Env* is encoded on a 4.2-kb single-spliced transcript, while *rex* and *tax* are encoded on a 2.1-kb double-spliced transcript. Unlike Tax, Rex regulates viral gene expression at the post-transcriptional level by influencing the ratio of unspliced to spliced mRNA exported to the cytoplasm. The effects of Rex are mediated by a specific 205 nucleotide *cis*-acting Rex response element (RRE) in the 3' region of the viral RNA. The RRE forms a stable and complex secondary structure, consisting of multiple stem loops, which is recognized by Rex. Early after infection, little Rex is available and double-spliced viral transcripts are efficiently transported to the cytoplasm to produce *rex/tax*

mRNA. Translation Tax greatly increases viral transcription. However, increased Rex expression exerts negative feedback control that decreases the export of double-spliced *rex/tax* mRNA in favour of unspliced and single-spliced mRNAs encoding viral structural proteins.

Adult T-cell Leukaemia (ATL)

HTLV-I infection is associated with the development of ATL, an aggressive, clonal malignancy of mature T lymphocytes, which develops in a small percentage of HTLV-I-infected individuals (Kondo *et al.*, 1987). Disease onset occurs after a long period of clinical latency, consistent with a multi-step process of T lymphocyte immortalization and transformation.

Leukaemic Cells

ATL is characterized by malignant proliferation of mature T lymphocytes with a phenotype of CD2⁺, CD3⁺, CD4⁺, CD8⁻, and HLA-DR⁺. Diagnostic criteria for ATL include incidental detection of leukocytosis, increased serum lactate dehydrogenase (LDH) levels, morphologically atypical lymphocytes with convoluted or lobulated nuclei, seropositivity for HTLV-I proteins, and detection of monoclonal or oligoclonal integrated provirus (Green *et al.*, 2001; Nicot, 2005). In addition, HTLV-I-transformed cells express surface markers associated with T-cell activation, including interleukin-2 receptor alpha (IL-2R α), transferrin receptor, and major histocompatibility complex class-II (MHC-II) molecules (Green *et al.*, 1989; Green *et al.*, 2001). HTLV-I-transformed cells also display increased expression of cellular genes that regulate T-cell growth and proliferation, including granulocyte-macrophage colony stimulating factor (GM-CSF), tumour necrosis factor-alpha (TNF- α), interleukin-15 (IL-15) and IL-2 (Azimi *et al.*, 1998; Ballard *et al.*, 1988; Green *et al.*, 2001; Mariner *et al.*, 2001; Uchiyama *et al.*, 1985). Although leukaemic cells from ATL patients express high levels of IL-2 and its receptor, these cells are mostly unresponsive to this cytokine and proliferate continuously in the absence of exogenous IL-2 (Maeda *et al.*, 1985; Popovic *et al.*, 1984).

Clinical Features

ATL can be classified into three clinical phases: acute, smouldering, and chronic phases. Smouldering and chronic phases of ATL are less aggressive than the acute phase of disease and are characterized by skin lesions and low levels of circulating lymphocytes. During chronic ATL, an increase in circulating leukaemic cells accounts for an increased leukocyte count. Patients in both the smouldering and chronic phases of ATL can progress into acute ATL within a period of months. In acute ATL, patients exhibit hypercalcaemia, lymphadenopathy, skin lesions due to leukaemic cell infiltration, leukocytosis, eosinophilia and neutrophilia, hepatosplenomegaly, hyperglycaemia, and renal failure. Expansion of a clonally infected CD4⁺ T lymphocyte population occurs during this phase (Eiraku *et al.*, 1998). Although the provirus remains

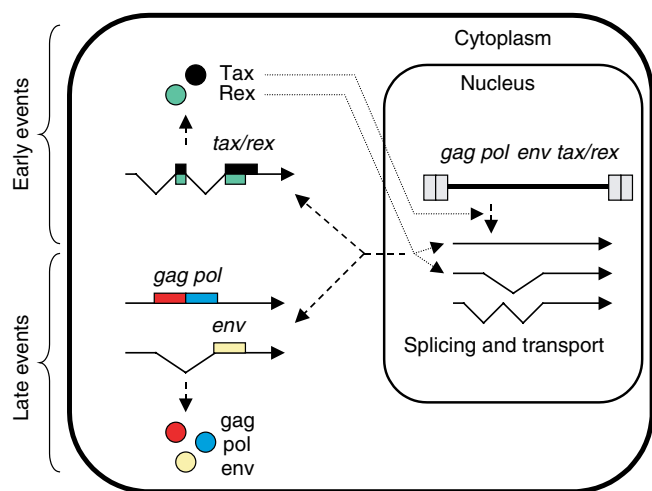


Figure 3 Transcriptional trans-activation and viral RNA processing.

integrated in the host genome, viral gene expression is not detectable. The prognosis of patients with acute ATL is poor, with a mean survival time of 6 months, following diagnosis (Green *et al.*, 2001; Nicot, 2005). Patients are typically infected early in life and disease onset occurs after a period of several decades (Wilks *et al.*, 1996). Opportunistic infections are a major complication due to the immunocompromized state of ATL patients. Common infections include *Pneumocystis carinii* pneumonia, cytomegalovirus pneumonia, aspergillosis, candidiasis, and *Strongyloides stercoralis* infection (Greenberg *et al.*, 1990; Newton *et al.*, 1992; Nicot, 2005; Pagliuca *et al.*, 1988; Roudier *et al.*, 1997; Tobinai *et al.*, 1991). Common opportunistic malignancies of ATL patients include Kaposi sarcoma and Epstein-Barr virus-associated lymphoma (Greenberg *et al.*, 1990; Tobinai *et al.*, 1991).

HTLV-I Tax and Mechanisms of Transformation

Unlike transducing or *cis*-acting retroviruses, HTLV-I does not encode a proto-oncogene and has not been associated with proviral insertion that activates the expression of flanking cellular oncogenes. Instead, extensive studies have demonstrated that the viral-encoded Tax protein, which is responsible for trans-activation of the viral promoter, is the major transforming protein. The role of Tax in leukaemogenesis was initially suggested by experiments in which rodent fibroblasts became morphologically transformed following Tax expression (reviewed in Feuer *et al.*, 2005). Subsequently, Tax was shown to induce tumours in transgenic animals, and immortalize human T-cells. For example, delivery of Tax by a transformation-defective *Herpes saimiri* virus vector immortalized primary human T lymphocytes *in vitro*, and these transformed cells resembled the leukaemic phenotype of HTLV-I-infected T-cells from patients (Grassmann *et al.*, 1989). Together, these studies have shown that Tax is sufficient for cellular transformation.

Interestingly, only a small percentage of leukaemic cells isolated from ATL patients express Tax. In these cells, Tax expression is low and typically can be detected only by PCR. Tax is expressed early after infection, before viral gene expression is downregulated by Rex-mediated suppression. As such, Tax is believed to play a role in the initiation of transformation. Whether Tax expression is required to maintain transformation remains to be determined. Anti-sense oligonucleotides designed to reduce Tax expression in tumour cell lines from Tax transgenic mice did not affect the rate of cell growth or tumour formation *in vivo*, despite a 90% reduction in Tax expression (Kitajima *et al.*, 1992). Thus, Tax expression may not be required to maintain the transformed state. However, HTLV-I-transformed Rat1 cells that no longer express Tax, lost the ability to form tumours in nude mice, and reintroduction of the *tax* gene restored the ability of these cells to form tumours, suggesting that Tax is required to both establish and maintain cellular transformation (Yamaoka *et al.*, 1992). Although there is no clear consensus, the cumulative results indicate that Tax may

contribute to the initiation of the transformation process but may not be required to maintain the transformed state.

Transcriptional Activation and Repression

One of the proposed mechanisms of Tax-mediated cellular transformation is transcriptional activation of cellular genes involved in proliferation and regulation of cellular processes. In addition to altering the DNA binding affinity and specificity of bZIP transcription factors, such as CREB, Tax also activates promoters that contain binding sites for the NF- κ B family of transcription factors (Figure 4). Tax activates the NF- κ B pathway by inducing I κ B Kinase-dependent phosphorylation of the inhibitor of NF- κ B, I κ B, causing it to release NF- κ B so that it can be translocated to the nucleus and can activate the cellular gene expression (Carter *et al.*, 2001; Suzuki *et al.*, 1995; Xiao *et al.*, 2000). As with CREB, Tax-mediated NF- κ B activation bypasses signalling pathways that normally regulate NF- κ B activity.

In addition to the CREB and NF- κ B pathways, Tax also activates genes regulated by the serum response factor (SRF) pathway. Genes with serum response elements (SREs) within their promoters are normally activated by the binding of SRF to a core DNA sequence, the CA₃G box, within the SRE. SRF then associates with ternary complex factors (TCFs) that bind to adjacent Ets-binding sites, also within the SRE. Tax has been shown to interact with both SRFs and TCFs to enhance their affinity for the SRE (Fujii *et al.*, 1992; Fujii *et al.*, 1995; Shuh *et al.*, 2000). Tax also affects SRE trans-activation due to its interaction with, and recruitment of, the coactivators CBP/p300 and p/CAF.

Despite being best known as a *transcriptional activator*, Tax can also repress transcription of cellular genes, in particular those involved in DNA repair and cell cycle checkpoint regulation. Transcriptional repression by Tax appears to be a consequence of some of the same mechanisms that induce transcriptional activation. For example, since Tax is known

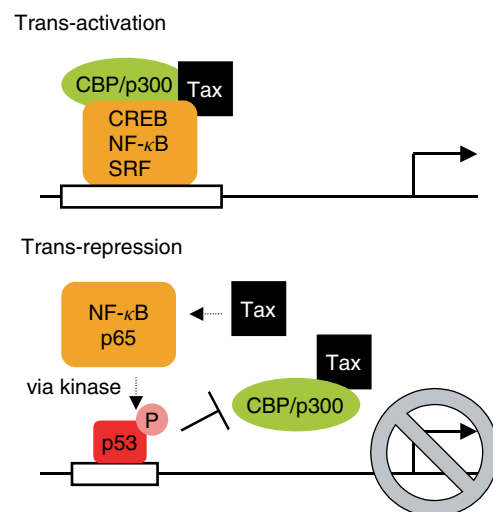


Figure 4 Tax-mediated transcriptional trans-activation or trans-repression of cellular genes.

to bind CBP/p300, it can sequester these important coactivators away from promoters that do not contain TREs, resulting in repressed expression of CBP/p300 dependent genes. For instance, Tax competes with E-box-binding members of the β helix-loop-helix transcription factor family for recruitment of CBP/p300 (Colgin *et al.*, 1998). DNA polymerase β , *lck*, *bax*, *p53*, *p18^{INK4c}*, and *p19^{INK4d}*, all of which contain E-box elements in their promoters, have been shown to be repressed by Tax (Akagi *et al.*, 1996; Suzuki *et al.*, 1999a; Suzuki *et al.*, 1999b). In addition, Tax inhibits p53-dependent transactivation of cellular promoters by competitively binding and sequestering CBP, which is required for p53-mediated transactivation. The repression of p53-mediated transcription by Tax is, however, more complex than simple sequestration of CBP/p300. Tax also represses p53-mediated transcription in an NF- κ B-dependent manner by inducing p53 phosphorylation, which alters the ability of p53 to interact with a variety of transcription factors (Jeong *et al.*, 2004). Tax-induced loss of p53 function inhibits G₁/S checkpoint arrest, p53-mediated apoptosis, and DNA repair.

Deregulation of Cell Cycle Checkpoint Controls

Separate from its effects on transcription regulation, Tax also interferes with normal regulation of cell cycle checkpoints via protein-protein interactions (Figure 5). During G₁ phase of the cell cycle, cyclin D forms an active complex with Cdk4 or Cdk6. The cyclin D/Cdk4/6 complex phosphorylates retinoblastoma (Rb), which, in its hypo-phosphorylated state, is bound to the transcription factor E2f. Rb phosphorylation releases E2f, enabling transcriptional activation of genes that promote cell cycle progression into S phase. Tax modulates this transition by two mechanisms. First, Tax directly interacts with cyclin D3, cyclin D2, Cdk4, and Cdk6, thereby stabilizing cyclin/Cdk complexes. This induces earlier onset of cyclin D/Cdk4/6 kinase activity and subsequent accumulation of hyper-phosphorylated Rb in Tax-expressing cells

(Haller *et al.*, 2002). Second, Tax binds directly to hypo-phosphorylated Rb to facilitate its proteasomal degradation (Kehn *et al.*, 2005). Through these mechanisms, the early release of E2f in Tax-expressing cells truncates G₁ phase, interferes with checkpoint control, and accelerates cell cycle entry into S phase (Lemoine *et al.*, 2001).

Repair of DNA damage prior to DNA synthesis in S phase is critical for accurate genome replication. Typically, activation of p53 in response to DNA damage regulates cell cycle progression between G₁ and S phases by delaying S phase entry and allowing time for DNA repair to occur prior to DNA replication. p53-dependent transcriptional activation of Cdk inhibitors (CKIs), such as p21^{Waf1/Cip1}, p27^{Kip1}, p15^{INK4b}, p16^{INK4a}, p18^{INK4c}, and p19^{INK4d}, blocks cyclin-Cdk kinase activities, resulting in cell cycle arrest at the G₁/S transition (Morgan, 1995; Peter *et al.*, 1994; Sherr *et al.*, 1995). The inactivation of p53 function by Tax interferes with these normal p53 activities, which allows cells to bypass the G₁/S checkpoint and enter the S phase where damaged DNA can be replicated.

The ability of Tax to activate expression of the Cdk inhibitor p21^{Waf1/Cip1} in a p53-independent manner appears to conflict with the inhibition of p53-dependent p21^{Waf1/Cip1} expression by Tax. However, p21^{Waf1/Cip1} is unique among the CKIs because it can bind to and stabilize cyclin-dependent kinase (CDK) complexes, such as cyclin D-Cdk4, and increase their kinase activity (Haller *et al.*, 2002). Therefore, Tax-mediated activation of p21^{Waf1/Cip1} expression increases the steady state levels of active cyclin D-Cdk4 and enhances Rb phosphorylation, allowing E2f release and activation of genes that promote cell cycle progression. Thus, even though p21^{Waf1/Cip1} is a cell cycle inhibitor, its activation by Tax may contribute to the accelerated progression of cells through G₁ phase and through the DNA damage-induced G₁/S checkpoint (Akagi *et al.*, 1996; Cereseto *et al.*, 1996; De La Fuente *et al.*, 2000; Hatta *et al.*, 2002).

Tax also interferes with the function of other CKIs. For example, the binding of Tax to the Cdk4/6 inhibitors, p15^{INK4b} and p16^{INK4a}, obstructs their ability to block kinase activity (Hatta *et al.*, 2002; Iwanaga *et al.*, 2001; Low *et al.*, 1997; Suzuki *et al.*, 1996). Tax also represses the transcription of p18^{INK4c} and p19^{INK4d} resulting in decreased activity of these cell cycle inhibitors (Suzuki *et al.*, 1999a). Together, these effects stimulate cell cycle progression by Tax.

DNA damage-induced cell cycle arrest can also occur at the G₂/M checkpoint. Following DNA damage, signalling cascades involving Chk1 and Chk2 activate several downstream targets including p53 (G₁ cell cycle arrest), Cdc25 (S phase delay), and Cdc25A/C (G₂ arrest). Phosphorylation of Cdc25A by Chk1 targets it for proteasomal degradation thereby inhibiting activation of the Cdk1/2 complex, which is required for progression through S phase and G₂/M checkpoints. Tax physically interacts with Chk1 and inhibits its ability to phosphorylate p53 (Park *et al.*, 2004). In addition, Cdc25A degradation is blocked in Tax-expressing cells, leading to attenuation of G₂ cell cycle arrest. Although Tax binds to Chk2, and this complex co-localizes at sites of DNA damage (Haoudi *et al.*, 2003), a function for this complex remains to be determined.

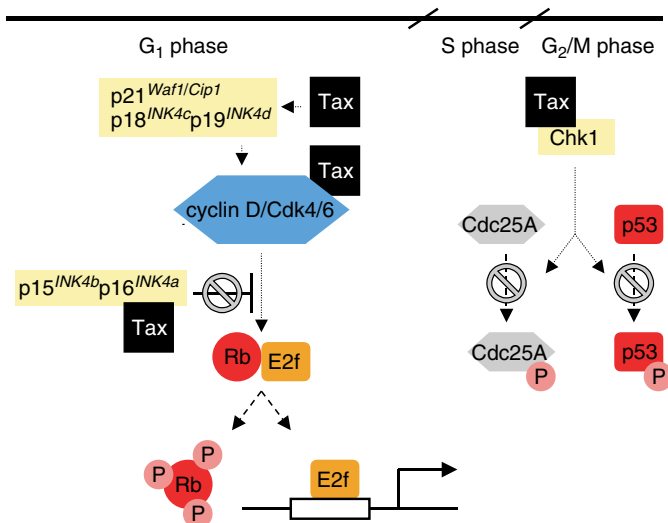


Figure 5 Tax-mediated dysregulation of cell cycle checkpoint controls.

Inhibition of DNA Repair and Promotion of Chromosome Instability

Cellular transformation is frequently associated with genome instability and HTLV-I-transformed lymphocytes isolated from ATL patients as well as Tax-transformed cells in culture display a wide range of chromosome abnormalities, including duplications, deletions, translocations, rearrangements, and aneuploidy (Whang-Peng *et al.*, 1985). Despite the complicated karyotypes observed in ATL patients, no specific chromosomal defect or mutation has been consistently associated with this leukaemia. Although Tax enacts pleiotropic dysregulation of cellular functions, there is no evidence that Tax directly induces DNA damage (Kao *et al.*, 1999). Instead, Tax appears to inhibit the ability of cells to repair DNA damage, resulting in an increased mutation frequency.

The cellular response to DNA damage is mediated through multiple functionally overlapping pathways, including nucleotide excision repair (NER), base excision repair (BER), mismatch repair (MMR), and double-strand break repair (DSBR). Tax suppresses some of these pathways, which likely increases the incidence of genomic mutations and contributes to Tax-mediated cellular transformation (Figure 6). For example, the ability of Tax to suppress NER correlates with its ability to trans-activate the expression of proliferating cell nuclear antigen (PCNA) (Ressler *et al.*, 1997; Kao *et al.*, 1999; Lemoine *et al.*, 2000). PCNA is a trimeric sliding clamp cofactor that increases the processivity of DNA polymerase δ during DNA replication and repair. PCNA overexpression, alone, is able to promote DNA synthesis past template lesions, resulting in increased nucleotide mis-incorporation by DNA polymerase δ (Mozzherin *et al.*, 1996; Mozzherin *et al.*, 1997). By increasing PCNA expression, Tax appears to reduce the fidelity of DNA polymerase δ during DNA replication and promote the introduction of mutations into the chromosomes of HTLV-I-infected cells. Tax also dysregulates DNA repair pathways by inhibiting BER, in part by repressing transcription of DNA polymerase β , an essential enzyme involved in base excision and mismatch forms of DNA repair (Jeang *et al.*, 1990). BER removes a variety of DNA lesions caused by spontaneous hydrolytic depurination, deamination of cytosine and 5-methylcytosine, products of reactions with hydroxyl free radicals, and covalent DNA adducts.

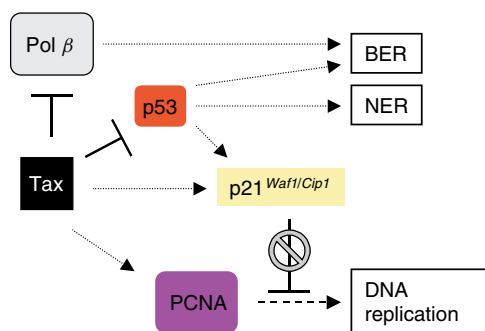


Figure 6 Tax-mediated inhibition of DNA repair and promotion of genomic instability.

The ability of Tax to suppress NER and BER, together with the abundance of chromosomal aberrations observed in Tax-expressing cells, suggests that Tax affects genome stability by multiple and, probably, overlapping mechanisms. Disruption of the mitotic spindle checkpoint (MSC) is associated with altered chromosome structures and numbers (Lavia *et al.*, 2003). The MSC regulates cell cycle transition from metaphase to anaphase and is mediated by numerous protein groups, including mitotic arrest defective (MAD-1, -2, -3) proteins, budding uninhibited by benzimidazoles (BUB-1, -2, -3), monopolar spindle (Mps-1), and anaphase promoting complex (APC) (Abrieu *et al.*, 2001; Cohen-Fix *et al.*, 1996; Hoyt *et al.*, 1991; Li *et al.*, 1991). The MSC is activated in response to DNA damage and, once activated, arrests the cell cycle prior to anaphase to ensure proper chromosome alignment before segregation. Malfunction of the MSC can result in improper sister chromatid separation and can lead to loss or gain of genetic material. Tax directly interacts with MAD-1 and APC to interfere with the activities of both proteins, resulting in multi-nucleation and delayed mitotic transition (Jin *et al.*, 1998; Liu *et al.*, 2005).

Dysregulation of human telomerase reverse transcriptase (hTERT) provides another opportunity to generate chromosomal abnormalities in Tax-expressing cells. hTERT adds specific nucleotide repeat sequences, known as *telomeres*, to the ends of chromosomal DNA. Telomeres prevent both the fusion of chromosomal ends and their degradation by exonucleases. In the absence of hTERT, telomere shortening occurs, allowing end-to-end chromosomal fusion and formation of dicentric chromosomes that are vulnerable to breakage during mitosis. This can lead to aneuploidy or chromosomal rearrangements and translocations. hTERT expression is elevated in most cancer cells, including leukaemic ATL cells (Franzese *et al.*, 2002; Tsumuki *et al.*, 2001; Uchida *et al.*, 1999). However, Tax represses hTERT expression during mitogenic stimulation, while activating hTERT expression in the absence of mitogenic stimulation (Gabet *et al.*, 2003; Sinha-Datta *et al.*, 2004). Reduced hTERT expression after T-cell stimulation can cause a transient state of genomic instability that is relieved when the mitogenic stimulus is removed. The biological advantage of hTERT repression for HTLV-I, if any, remains to be determined. However, activation of hTERT by Tax upon removal of the mitogenic signal may offer a long-term proliferative advantage to cells that have acquired chromosomal abnormalities.

CONCLUSION

For several decades, RNA tumour viruses, including HTLV-I, have been intensely studied because of their association with human cancers and their usefulness as model systems in which to study processes of cellular transformation. The HTLV-I Tax oncoprotein contributes to tumourigenesis by interfering with cellular transcription regulation, tumour suppressor function, cell cycle regulation, and DNA repair (Figure 7). Alteration of the host environment appears to be an incidental result of viral genome replication efforts

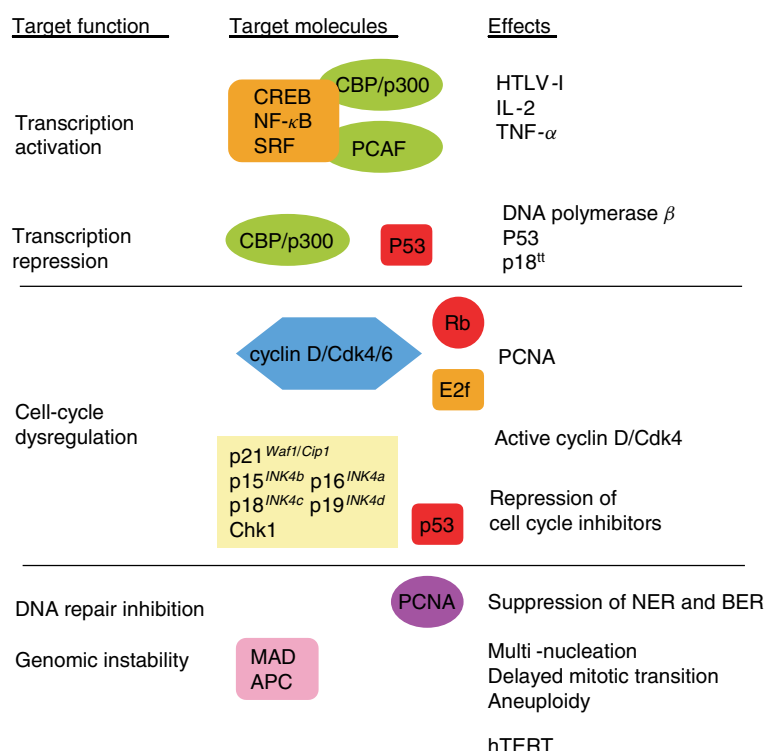


Figure 7 Pleiotropic effects of Tax.

that indirectly allow an accumulation of somatic mutations and genomic instability. It is almost certain that continued study of HTLV-I-mediated transformation will provide new insight into the roles of viral and cellular factors in transformation and may reveal new basic cellular mechanisms of carcinogenesis.

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